

Project Outline

A sufficient and sustainable treatment for borderline personality disorder (BPD) has yet to be discovered—using current treatments such as dialectical behaviour therapy, diagnostic remission occurs in 85% to 93% of patients over 10 years (Soloff, 2021). While remission rates are high, patients experience limited long-term recovery, highlighting the opportunity for more effective and lasting treatments. Experiments on mice have indicated that optogenetics may be used to elicit specific memories, induce false ones, and neutralize fear associated with a specific memory (Liu et al., 2012). Although research on optogenetics for psychiatric treatment in humans is limited, studies using optogenetics for PTSD and fear extinction suggests a possible pathway forward for mood and personality disorders. However, research about optogenetics on human participants for psychiatric purposes is sparse, and has yet to be explored extensively due to its potentially problematic nature. Because optogenetics target the neural networks linked to emotional memory, this intervention could potentially be more durable than traditional therapies that rely on behaviour alone. Our research project raises the following question: To what extent can optogenetics be utilized to address BPD symptoms in humans?

Optogenetics is an invasive procedure that raises both practical and ethical concerns (Goshen, 2014). Functionally, it may be difficult to isolate a single, specific memory (Kohara & Okada, 2023). Ethical questions about memory manipulation and personal autonomy also arise. Our project will discuss the functional probability of current day optogenetics on humans, as well as the necessary ethical considerations.

The experiment will take a voluntary sample of participants diagnosed with BPD based on the DSM-5-TR criteria. Recruitment will prioritize individuals with treatment-resistant BPD who are at risk of suicide, to evaluate the potential of optogenetics as an alternative treatment. Participants will be randomly assigned to either a control group (receiving a placebo treatment with no changes to their memory) or an experimental group (receiving optogenetic treatment). BPD symptoms measured before and after treatments through clinical interviews and self-report scales (eg. BPDSI, BSL-23) will determine if the experimental group shows a statistically significant reduction in BPD symptoms. These assessment choices were chosen as they were deemed sufficient in measuring BPD (Arntz et al., 2003; Bohus, 2009). Prior to the experiment, the specific memory of interest will be determined based on clinical interviews for the experimental group.

We foresee the results contributing to psychology in examining how BPD works in the human brain, and explore a potential new treatment for humans using optogenetics. Moreover, contributions to philosophy include the implications of tampering with the nature of selfhood and the extent to which memories make up human identity. The combination of these contributions will provide greater knowledge on BPD intervention.